SOFT3111 – Internet and Web Programming

Project #1: Web Application Project Proposal

StudyBuddy – AI-Powered Course Resource and Collaborative Study Platform

2. Required Proposal Sections

2.1. Project Information

Project title: StudyBuddy – AI-Powered Course Resource and Collaborative Study Platform

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Course: SOFT3111 – Internet and Web Programming

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GitHub repository URL: https://github.com/ecenurdemirx/SOFT3111-Group08-StudyBuddy.git

2.2. Project Overview and Motivation

StudyBuddy is an AI-powered collaborative learning web application designed for university students. The platform organizes academic materials by university course and topic while helping students connect with classmates and study together. Registered students can share PDF lecture notes, slides, worksheets, practice questions and past exam papers. Rather than building interactive question banks or online quizzes, StudyBuddy focuses on making existing documents easy to upload, discover, preview and download.

University learning resources are often fragmented across messaging groups, personal drives and individual devices. Students may miss valuable materials simply because they are not members of a particular chat. Even after finding a document, they may have trouble understanding a difficult topic or solving a question. Students taking the same course also struggle to find suitable study partners or organize a face-to-face group study session.

StudyBuddy solves these connected problems through a course-centered resource library, student connections and AI-assisted learning. Students can filter materials by course, topic, document type and academic term; bookmark and report resources; and send friend requests to other students. They can create study meetups linked to a course, specifying a topic, location, date, time and capacity, while classmates can request to join. The platform

supports in-person study coordination without requiring an advanced real-time chat system.

An AI Study Assistant provides an additional learning layer. Students may request summaries of uploaded PDF materials, ask for a step-by-step explanation of a question from a document, or receive a suggested study plan for a particular course or exam. Where possible, explanations will reference the selected document, and the interface will warn users that AI responses may contain errors. The core AI features will use a server-side integration with an external AI service rather than requiring the group to train a model.

The main beneficiaries are students who need accessible academic resources, students seeking peers to study with, and moderators who help maintain a reliable collection. By connecting course materials, real-world collaboration and guided AI support in one place, StudyBuddy promotes a more organized and cooperative study experience. The project will demonstrate ASP.NET MVC, C#, SQL database operations, HTML/CSS/JavaScript, Ajax and JSON through a practical semester-scale web application.

2.3. Target Users and User Roles

Guest – A visitor who wants to explore available courses and public resource details. Guests may browse courses and resource titles but must register to access PDFs, AI assistance, friendships and study meetups.

Student – The primary registered user. Students upload and browse PDF resources; search by course or topic; preview, download and bookmark materials; add friends; organize and join study meetups; and use AI-based summaries, question explanations and study guidance.

Moderator – A trusted student or course representative responsible for reviewing newly submitted or reported documents and managing inappropriate posts or meetup listings in assigned courses.

Administrator – A team member responsible for managing course and topic catalogs, user accounts, moderator assignments and overall platform settings. Administrative access is checked independently of normal student permissions.

2.4. Functional Requirements

High-priority requirements define the minimum viable application. Medium-priority items will be developed after the main workflow; Low-priority items are optional enhancements.

ID	Functional Requirement	Priority

FR-01	Users can register, sign in and sign out; protected features require authentication.	High
FR-02	Students can upload a PDF (maximum 20 MB) with title, description, course, topic, document type and academic term.	High
FR-03	Students can browse courses and topics; preview or download approved PDF notes, worksheets, practice-question files and past exam papers.	High
FR-04	Students can search resources by keyword and filter by course, topic, document type and term without a full page reload (Ajax).	High
FR-05	Students can send, accept or decline friend requests and see their confirmed friends.	High
FR-06	Students can create an in-person study meetup for a course with topic, location, date, time and participant capacity.	High
FR-07	Students can browse study meetups related to their courses, request to join and see whether their participation is approved. Organizers can accept or decline requests.	High
FR-08	Students can request an AI-generated summary of the text from a selected uploaded PDF; the application identifies the source document.	High
FR-09	Students can select or paste a question from a PDF and request an AI explanation with reasoning steps rather than an interactive quiz.	High
FR-10	New uploads remain pending until moderator approval; moderators can approve or reject	High

	submitted resources and remove reported items.	
FR-11	Students can bookmark documents and view their saved resources.	Medium
FR-12	Students can request an AI study plan based on a course, selected topics, available time and target exam date.	Medium
FR-13	Students can report inappropriate or inaccurate resources and meetup listings with a reason.	Medium
FR-14	Administrators can create and edit course/topic records, manage accounts and assign moderator permissions.	Medium
FR-15	Students can add short comments or ratings to shared resources.	Low

2.5. Non-Functional Requirements

NFR-01 Security: Passwords must be stored as salted hashes, and server-side authorization must protect document access, moderator actions and administrator pages. File types and sizes are validated on the server.

NFR-02 Privacy: Friend requests and study meetup participation information are shown only to authorized users; personal details are not exposed to guests.

NFR-03 Responsiveness: Pages must display without horizontal scrolling on mobile screens of 360 px width and desktop screens up to 1920 px.

NFR-04 Performance: Search/filter responses should complete in under 3 seconds with 1,000 indexed resources in a development test environment; AI requests show a loading state.

NFR-05 Reliability: Invalid PDFs and missing required fields produce actionable error messages; failed AI requests do not prevent normal resource access and show a retry option.

NFR-06 AI transparency: AI output is labeled AI-generated, references the selected source document when used, and warns that explanations may be incorrect. Text is sent to the AI provider only for an explicitly requested AI action.

NFR-07 Compatibility and accessibility: Core features must work on current Chrome, Edge, Firefox and Safari, with labeled form controls and keyboard-accessible navigation.

2.6. User Scenarios

Scenario 1: Share a Past Exam PDF

Actor: Student

Preconditions: Student is signed in; course and topic exist.

Main flow: 1. Student selects the course and topic. 2. Student chooses Upload Resource and selects a PDF of past exam questions. 3. Student fills in title, type and term. 4. System validates the file and saves it as Pending Approval. 5. Moderator approves it.

Expected outcome: The PDF appears in the appropriate course/topic listing for other students to preview or download.

Related requirements: FR-02, FR-03, FR-10

Scenario 2: Arrange a Group Study Meetup

Actor: Student

Preconditions: Student is signed in; relevant course exists.

Main flow: 1. Student opens a course page. 2. Student creates a meetup for a specific topic with date, time, place and capacity. 3. Another student finds the meetup and requests to join. 4. Organizer accepts the request. 5. Participants view the confirmed meetup details.

Expected outcome: Students taking the same course can coordinate an in-person study session.

Related requirements: FR-05, FR-06, FR-07

Scenario 3: Understand a Difficult PDF Question with AI

Actor: Student

Preconditions: An approved PDF is accessible; the AI service is available.

Main flow: 1. Student opens a past exam PDF. 2. Student selects or copies a difficult question. 3. Student clicks Explain with AI. 4. System sends the question and relevant extracted PDF text to the AI service. 5. The response provides an explanation and identifies it as AI-generated.

Expected outcome: Student receives guided assistance without completing an in-app quiz.

Related requirements: FR-03, FR-09

Scenario 4: Summarize Notes and Plan Revision

Actor: Student

Preconditions: Student is signed in; an approved text-based PDF is available.

Main flow: 1. Student opens lecture notes and requests an AI summary. 2. System extracts document text and returns a condensed explanation. 3. Student opens the Study Plan tool, specifies topics, free time and exam date. 4. System generates a suggested revision schedule.

Expected outcome: Student can understand the material faster and organize preparation.

Related requirements: FR-08, FR-12